

Signed-Kernel Oscillation at Every Depth

Minimal Pick Compensation, Zero Geometry,
and Certified Polylogarithmic Identities

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Abstract

We extend the manuscript's depth-two signed-kernel method to strict single-variable multiple polylogarithms of arbitrary depth. The first $d - 1$ indices are positive integers and the final index may be any positive real number. Two bounded operators on $L^1(0, 1)$ construct a density with exactly $d - 1$ simple sign changes, $d - 1$ vanishing initial moments, and norm at most 2^{d-1} . The sign-change polynomial converts this signed measure into a positive measure and yields a canonical factorization $L_s(z) = z^d H_s(z) / \prod_j (1 - r_j z)$. We prove that this polynomial compensation is of minimal degree, that the principal continuation has no nonzero zero on $\mathbb{C} \setminus [1, \infty)$, and that its imaginary part has exactly $d - 1$ simple zeros on every upper semicircle of radius at most one. A direct confluent Cauchy-kernel argument proves the upper zero bound; a positive-transform argument supplies the matching lower bound, including the singular unit-radius endpoint.

Further results include strict inward angular motion, interlacing of density zeros, small-radius and large-outer-index expansions, positive terminal mixtures, a shifted-terminal Lerch extension, explicit logistic-root identities for the all-ones family, and rational error certificates at Gaussian points. The package contains 24 certified angular-root brackets and 5,713 finite exact check groups. The proofs are ordinary analytic proofs, not proof-assistant formalizations. The S_6 and S_8 arithmetic conjectures, unrestricted fractional leading indices, and the existing normalized-radius conjecture are not claimed solved. No universal claim of literature priority is made.

Repository reference. <https://github.com/VladimirReshetnikov/ProveIt>

Reviewed snapshot: 3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f. This is an additive research package; it does not alter the repository or replace its canonical manuscript.

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Reading map. Sections 2–5 contain the complete structural proof. Section 6 derives asymptotic consequences, Section 7 gives explicit identities, and Section 8 explains the exact certificates. Sections 9–10 separate corrections, scope limits, and future research. The appendices record independent arithmetic controls and integration dependencies.

1 The problem, conventions, and contribution

The existing ProveIt manuscript develops a successful bridge between nested sums, signed Hausdorff moments, positive transforms, and angular zero geometry. Its depth-two function is

$$F_{a,b}(z) = \sum_{n \geq 1} \frac{H_{n-1}^{(b)}}{n^a} z^n, \quad H_{n-1}^{(b)} = \sum_{m=1}^{n-1} m^{-b}.$$

In the reviewed snapshot, slit-plane nonvanishing and upper-semicircle angular uniqueness are already proved at every positive inner order and nonnegative outer order. The integer-order signed density and the later real-order and subcritical extensions are distinct parts of that argument [1, 2, 3]. We do not present these depth-two conclusions as newly resolved conjectures.

The question pursued here is what survives at arbitrary *strict* depth. A positive integral with a product of several resolvent factors is useful but does not, by itself, imply nonvanishing. The missing structure is a signed density whose number of sign changes matches exactly the number of initially vanishing coefficients. This equality allows polynomial compensation and controls the full angular zero count.

1.1 Strict single-variable notation

For an index of depth d , define

$$L_{\mathbf{s}}(z) = \text{Li}_{s_1, \dots, s_d}(z, 1, \dots, 1) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \dots n_d^{s_d}} = \sum_{n \geq 1} c_{\mathbf{s}}(n) z^n, \quad |z| < 1. \quad (1.1)$$

Our parameter domain is

$$s_1, \dots, s_{d-1} \in \mathbb{N}_{\geq 1}, \quad s_d = b > 0. \quad (1.2)$$

At depth one, this means any $b > 0$. Indices are strictly decreasing, not weakly decreasing. Only the outer summation variable receives z^{n_1} . This convention agrees with the single-variable convention in the manuscript and in the multiple-polylogarithm literature. Classical polylogarithm branch conventions are recorded in DLMF [8].

Write

$$C_{\mathbf{s}} = c_{\mathbf{s}}(d) = \frac{1}{d^{s_1} (d-1)^{s_2} \dots 2^{s_{d-1}}} > 0. \quad (1.3)$$

The empty product at depth one is one. The value of $C_{\mathbf{s}}$ is independent of b , because the innermost index of the first admissible chain is one. Also $c_{\mathbf{s}}(1) = \dots = c_{\mathbf{s}}(d-1) = 0$. Throughout, principal continuation means continuation to

$$\Omega = \mathbb{C} \setminus [1, \infty).$$

The word “zero” refers to a zero of a holomorphic function, unless the imaginary part or a signed density is explicitly specified.

1.2 Main result

Theorem 1.1 (All-depth compensation and zero geometry). *Let (1.2) hold. There is a smooth real density $\kappa_{\mathbf{s}} \in L^1(0, 1)$ such that*

$$\int_0^1 u^{n-1} \kappa_{\mathbf{s}}(u) du = c_{\mathbf{s}}(n), \quad \|\kappa_{\mathbf{s}}\|_1 \leq 2^{d-1}.$$

It has exactly $d-1$ simple zeros $0 < r_1 < \dots < r_{d-1} < 1$, with alternating signs and positive sign near one. Set

$$P_{\mathbf{s}}(u) = \prod_{j=1}^{d-1} (u - r_j), \quad D_{\mathbf{s}}(z) = \prod_{j=1}^{d-1} (1 - r_j z).$$

Then $d\mu_s = P_s \kappa_s du$ is a positive measure of mass C_s , and

$$L_s(z) = \frac{z^d}{D_s(z)} H_s(z), \quad H_s(z) = \int_0^1 \frac{d\mu_s(u)}{1-zu}. \quad (1.4)$$

The polynomial D_s is the unique normalized polynomial of smallest degree making $D_s L_s / z^d$ a positive Hausdorff–Stieltjes transform. In particular, L_s has no zero in Ω other than its zero of multiplicity d at zero.

For every $0 < \rho \leq 1$, there are exactly $d - 1$ simple zeros

$$0 < \theta_1(\rho) < \cdots < \theta_{d-1}(\rho) < \pi$$

of $\theta \mapsto \operatorname{Im} L_s(\rho e^{i\theta})$. With $\alpha_k = k\pi/d$ they satisfy

$$\alpha_k - \arcsin(\rho \sin \alpha_k) < \theta_k(\rho) < \alpha_k. \quad (1.5)$$

Every branch satisfies $\theta'_k(\rho) < 0$ for $0 < \rho \leq 1$ (with the one-sided derivative at one). The imaginary part starts positive and alternates sign at these zeros; the value of L_s at the k th crossing is real with sign $(-1)^k$.

The proof is divided into operator, oscillation, compensation, and angular-count steps below. None of those steps is inferred from the finite verification suite.

1.3 Relation to prior work and what is new here

Positive Hausdorff generating functions have a classical Pick-function interpretation; Liu and Pego give a systematic treatment [6]. The present H_s belongs to that class. The substantial step is not that a positive measure has a Pick transform, but the construction of the *minimal* polynomial that changes a strict-depth signed measure into a positive one.

Recent work on multiple *star* polylogarithms also uses positive Hausdorff measures [7]. A star sum has weak inequalities in its nested indices. It must not be substituted for (1.1): its first nonzero coefficient, normalization, and positive-measure properties are different. Our proof treats strict sums directly and retains their $d - 1$ vanishing initial moments.

Relative to the reviewed manuscript sections, the added results are the two-operator all-depth construction, exact oscillation and interlacing, minimal positive compensation, the all-depth slit-plane and angular-zero theorems, their uniform asymptotic consequences, and explicit compensated integral identities. The elementary all-ones logarithm formula and the general positive-transform theory are background, not claims of discovery. A targeted literature search did not establish priority for every theorem stated here; a broader novelty review remains appropriate before publication.

Claim	Status and scope
All-depth kernel and zero count	Proved here on (1.2).
Canonical compensator	Proved minimal and unique among normalized least-degree polynomials.
Explicit integral identities	Proved; the underlying all-ones polylogarithm is classical.
Gaussian and root certificates	Finite exact rational checks with analytic error bounds.
Arbitrary fractional leading orders	Not covered by this proof.
S_6 , new S_8 , period independence	Not settled or promoted by this package.

2 Two bounded operators on signed moment densities

For $f \in L^1(0, 1)$ define its moments by

$$m_n(f) = \int_0^1 u^{n-1} f(u) du, \quad n \geq 1.$$

We use the two operators

$$(\mathbf{A}f)(u) = \int_u^1 \frac{f(v)}{v} dv, \quad (2.1)$$

$$(\mathbf{B}f)(u) = \int_0^u \frac{f(v)}{1-v} dv - \int_u^1 \frac{f(v)}{v} dv. \quad (2.2)$$

For every fixed $0 < u < 1$, both integrals in (2.2) are absolutely convergent. In particular, this definition never subtracts two divergent endpoint integrals. The operator $\mathbf{A}$ is the outer-index-raising operator already present at depth two in the manuscript [1]. The companion $\mathbf{B}$ encodes a strict prefix sum of moments.

Lemma 2.1 (Norm, derivative, and moment identities). *Both operators map $L^1(0, 1)$ to itself and satisfy*

$$\|\mathbf{A}f\|_1 \leq \|f\|_1, \quad \|\mathbf{B}f\|_1 \leq 2\|f\|_1. \quad (2.3)$$

Their moments are

$$m_n(\mathbf{A}f) = \frac{m_n(f)}{n}, \quad m_n(\mathbf{B}f) = \frac{1}{n} \sum_{j=1}^{n-1} m_j(f). \quad (2.4)$$

If f is continuous in the open interval, then

$$(\mathbf{A}f)'(u) = -\frac{f(u)}{u}, \quad (\mathbf{B}f)'(u) = \frac{f(u)}{u(1-u)}. \quad (2.5)$$

Moreover $\mathbf{A}f$ extends continuously to one with value zero, and $m_1(\mathbf{B}f) = 0$.

Proof. Tonelli's theorem applied to absolute values gives

$$\int_0^1 \int_u^1 \frac{|f(v)|}{v} dv du = \int_0^1 |f(v)| dv.$$

Similarly,

$$\int_0^1 \int_0^u \frac{|f(v)|}{1-v} dv du = \int_0^1 |f(v)| dv.$$

These identities prove the two operator bounds and justify every following application of Fubini to a signed integrand. For the first operator,

$$m_n(\mathbf{A}f) = \int_0^1 \frac{f(v)}{v} \left(\int_0^v u^{n-1} du \right) dv = \frac{m_n(f)}{n}.$$

The moment of the first term of $\mathbf{B}f$ is

$$\frac{1}{n} \int_0^1 \frac{1-v^n}{1-v} f(v) dv = \frac{1}{n} \sum_{j=1}^n m_j(f).$$

Subtracting $m_n(f)/n$ yields the strict prefix in (2.4). The derivative formulas follow locally from the fundamental theorem of calculus. Finally, $\int_u^1 f(v)/v dv \rightarrow 0$ as $u \uparrow 1$, because $1/v$ is bounded near one and f is integrable there. $\square$

2.1 Construction at arbitrary depth

The positive starting density is

$$f_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)}, \quad b > 0. \quad (2.6)$$

Its mass is one and the Gamma integral gives $m_n(f_b) = n^{-b}$. Define, with composition read from right to left,

$$\kappa_s = \mathbf{A}^{s_1-1} \mathbf{B} \mathbf{A}^{s_2-1} \mathbf{B} \cdots \mathbf{A}^{s_{d-1}-1} \mathbf{B} f_b. \quad (2.7)$$

There are $d-1$ occurrences of $\mathbf{B}$; all powers of $\mathbf{A}$ are nonnegative integer powers. At depth one the right side is simply f_b .

Proposition 2.2 (Moment realization and continuation). *The density (2.7) is smooth on $(0, 1)$, belongs to $L^1(0, 1)$, and satisfies*

$$m_n(\kappa_s) = c_s(n), \quad \|\kappa_s\|_1 \leq 2^{d-1}. \quad (2.8)$$

The principal continuation is

$$L_s(z) = z \int_0^1 \frac{\kappa_s(u)}{1 - zu} du, \quad z \in \Omega. \quad (2.9)$$

The density is the unique finite signed measure realizing these moments.

Proof. The coefficient recurrence for a nonempty tail $\mathbf{t}$ is

$$c_{(a, \mathbf{t})}(n) = \frac{1}{n^a} \sum_{j=1}^{n-1} c_{\mathbf{t}}(j).$$

It is exactly the effect of $A^{a-1}B$ in (2.4). Induction proves the moments and the norm bound. The derivative identities prove local smoothness, starting from (2.6). Expanding the denominator in the open disk is justified by the integrable bound $|\kappa|/(1 - |z|)$, and gives the series (1.1).

On a compact subset of Ω , $|1 - zu|$ is bounded away from zero uniformly in $u \in [0, 1]$. The signed integral is consequently holomorphic there and continues the same germ. If two finite signed measures have these moments, their difference integrates every polynomial to zero. Uniform polynomial approximation on $[0, 1]$, followed by uniqueness of a finite measure from its continuous-function integrals, proves equality. $\square$

Corollary 2.3 (Ordinary convergence on the nontrivial unit circle). *For (1.2), $c_s(n) \rightarrow 0$ and*

$$\sum_{n \geq 1} |c_s(n+1) - c_s(n)| \leq \|\kappa_s\|_1 \leq 2^{d-1}. \quad (2.10)$$

Thus the original series (1.1) converges at every $|z| = 1$, $z \neq 1$, and agrees there with (2.9).

Proof. Dominated convergence proves the coefficient limit. Also

$$|c(n+1) - c(n)| \leq \int_0^1 u^{n-1} (1-u) |\kappa(u)| du.$$

Summing the positive majorants gives (2.10). Summation by parts against the bounded partial sums of z^n then proves convergence when $|z| = 1$, $z \neq 1$. Abel's limiting theorem identifies its sum with the radial limit of the disk series, which in turn equals the continuous signed integral at that point. $\square$

This conclusion concerns the parameter domain of this article. It is not a claim that all of the manuscript's more general fractional-order boundary series converge ordinarily.

3 Exact oscillation and two interlacing laws

The central mechanism is a balance between moment orthogonality and Rolle's theorem. Vanishing moments force sign changes; the two derivative identities prevent extra zeros. This proof does not assume a variation-diminishing theorem for fractional integration.

Lemma 3.1 (Vanishing moments force sign changes). *Suppose a nonzero continuous real $f \in L^1(0, 1)$ has finitely many zeros and*

$$\int_0^1 u^j f(u) du = 0 \quad (0 \leq j \leq m-1).$$

Then f has at least m sign changes in $(0, 1)$.

Proof. If its sign changes were $t_1, \dots, t_k$ with $k < m$, the polynomial $Q(u) = \prod_{j=1}^k (u - t_j)$ would have degree at most $m - 1$ and Qf would have one fixed nonzero sign except at finitely many points. Its integral would be nonzero, contradicting orthogonality. A zero at which no sign change occurs is not included in Q . $\square$

Theorem 3.2 (Exact density zeros). *For (1.2), κ_s has exactly $d - 1$ distinct simple zeros in $(0, 1)$. Its sign alternates at those zeros, and it is positive on the interval to the right of its largest zero.*

Proof. At depth one, f_b is strictly positive and has no zero. Suppose first that a density f at depth $d - 1$ has exactly $d - 2$ simple zeros. By (2.5), the derivative of Bf has exactly those zeros. Rolle's theorem therefore bounds the number of distinct zeros of Bf by $d - 1$. This also rules out an infinite set of zeros: any d distinct zeros would already force too many zeros of the derivative.

The first $d - 1$ moments of the new density vanish, by (2.4). Lemma 3.1 supplies at least $d - 1$ sign changes, so the upper bound is attained. There is one derivative zero between each pair of consecutive new zeros, using all $d - 2$ derivative zeros. None remains at a new zero. Hence every new zero is simple.

Now suppose f is a depth- d density with exactly $d - 1$ simple zeros, and put $g = Af$. We have $g(1) = 0$. If g had k interior zeros, Rolle's theorem between successive interior zeros and between the largest one and the endpoint one would give at least k zeros of $g' = -f/u$. Thus $k \leq d - 1$. Its first $d - 1$ moments still vanish, so the same lower bound gives exactly $d - 1$ sign changes. The derivative zeros are now all allocated to those successive intervals, including the last interval ending at one. Consequently the interior zeros of g are simple. Repeating this argument for each occurrence of A and B proves the zero count.

Let $P(u)$ be the monic polynomial whose roots are the resulting $d - 1$ zeros. The product $P\kappa_s$ has a fixed strict sign off its roots. Orthogonality to degrees at most $d - 2$ gives

$$\int_0^1 P(u)\kappa_s(u) du = \int_0^1 u^{d-1}\kappa_s(u) du = C_s > 0.$$

Thus $P\kappa_s$ is positive, and since P is positive to the right of its largest root, so is κ_s . $\square$

Corollary 3.3 (Index raising and depth raising interlace). *If $f = \kappa_s$ has roots $r_1 < \dots < r_{d-1}$ and Af has roots $q_1 < \dots < q_{d-1}$, then*

$$q_1 < r_1 < q_2 < r_2 < \dots < q_{d-1} < r_{d-1} < 1. \quad (3.1)$$

Thus raising the first index by one moves every density zero strictly toward zero. If a depth- $(d - 1)$ density has roots $t_1 < \dots < t_{d-2}$ and Bf has roots $r_1 < \dots < r_{d-1}$, then

$$r_1 < t_1 < r_2 < \dots < t_{d-2} < r_{d-1}. \quad (3.2)$$

Proof. These are precisely the derivative-zero allocations in the preceding proof. In the A case the last interval is $(q_{d-1}, 1)$; in the B case all intervals are between consecutive interior zeros. The number of derivative zeros equals the number of allocated intervals, so the inequalities are strict and exhaustive. $\square$

3.1 Positive seeds, terminal mixtures, and a Lerch extension

The preceding proof used only positivity, smoothness, integrability, and the positive mass of the initial density. The explicit form (2.6) was needed to identify its moments as n^{-b} .

Theorem 3.4 (Positive-seed extension). *Let f be a smooth strictly positive density in $L^1(0, 1)$, of mass $M_0 > 0$. Replace f_b by f in (2.7). All density-zero, interlacing, compensation, slit-plane nonvanishing, and angular-count conclusions of this article remain valid. The norm bound becomes $2^{d-1}M_0$, and the mass of the positive compensated measure is*

$$\frac{M_0}{d^{s_1}(d-1)^{s_2} \dots 2^{s_{d-1}}}.$$

Proof. The operator identities still hold. After $d - 1$ strict prefix steps, the first $d - 1$ moments vanish and the first nonzero moment is the stated positive number. The proof of Theorem 3.2 therefore applies unchanged. The arguments in the next two sections use only these moment and oscillation properties, and so establish the remaining assertions as well. $\square$

In particular, for fixed leading indices, every finite positive linear combination

$$\sum_{j=1}^N a_j L_{s_1, \dots, s_{d-1}, b_j}(z), \quad a_j > 0, \quad b_j > 0, \quad (3.3)$$

has the same nonvanishing and exact angular-count properties. Its seed is $\sum a_j f_{b_j}$. This is a special convex-closure statement; sums of arbitrary zero-free holomorphic functions need not be zero-free.

For a second application, take $\lambda > -1$ and

$$f_{b, \lambda}(u) = u^\lambda \frac{(-\log u)^{b-1}}{\Gamma(b)}.$$

Its moments are $(n + \lambda)^{-b}$ and its mass is $(1 + \lambda)^{-b}$. The resulting strict shifted-terminal family is

$$L_{s_1, \dots, s_{d-1}, b}^{(\lambda)}(z) = \sum_{n_1 > \dots > n_d \geq 1} \frac{z^{n_1}}{n_1^{s_1} \dots n_{d-1}^{s_{d-1}} (n_d + \lambda)^b}. \quad (3.4)$$

It has the same minimal compensator degree $d - 1$ and exactly $d - 1$ angular zeros for $0 < \rho \leq 1$. At depth one it is the usual z -multiple of a Lerch series. No assertion is made here about shifts in all denominators simultaneously.

4 Canonical positive compensation and slit-plane nonvanishing

Fix $\mathbf{s}$ and abbreviate $\kappa = \kappa_{\mathbf{s}}$, $P = P_{\mathbf{s}}$, $D = D_{\mathbf{s}}$, and $C = C_{\mathbf{s}}$. Empty products at depth one are one. The preceding section proves that

$$P(u)\kappa(u) > 0 \quad \text{for all } u \in (0, 1) \setminus \{r_1, \dots, r_{d-1}\}. \quad (4.1)$$

Since P is bounded on $[0, 1]$, the positive measure $d\mu = P\kappa du$ is finite.

Lemma 4.1 (Polynomial cancellation). *Suppose $m_1(\kappa) = \dots = m_{d-1}(\kappa) = 0$. For any real polynomial E of degree at most $d - 1$, the removable quotient at zero satisfies*

$$E(z) \frac{L(z)}{z^d} = \int_0^1 \frac{u^{d-1} E(1/u) \kappa(u)}{1 - zu} du. \quad (4.2)$$

Here $u^{d-1} E(1/u)$ is a polynomial, including at $u = 0$.

Proof. First subtract the initial $d - 1$ zero moments in (2.9) to obtain

$$L(z) = z^d \int_0^1 \frac{u^{d-1} \kappa(u)}{1 - zu} du.$$

For $0 \leq j \leq d - 1$, multiplying this last transform by z^j and moving the factor z^j to the density changes it to $u^{d-1-j} \kappa(u)$. The difference is a finite linear combination of the moments of degrees $d - 1 - j, \dots, d - 2$, all zero. Applying this monomial identity to every coefficient of E proves (4.2) in the disk and hence throughout Ω . $\square$

Theorem 4.2 (Positive factorization and nonvanishing). *The measure μ has mass C , and*

$$D(z)L(z) = z^d \int_0^1 \frac{d\mu(u)}{1-zu}. \quad (4.3)$$

Its transform H has strictly positive imaginary part when $\operatorname{Im} z > 0$, strictly negative imaginary part when $\operatorname{Im} z < 0$, and $H(x) > 0$ for every real $x < 1$. The only zero of L on Ω is its zero of multiplicity d at zero.

Proof. We have $u^{d-1}D(1/u) = P(u)$, so the preceding lemma gives (4.3). Its value after division by z^d at zero is $\int P\kappa = C$, as already shown in the oscillation proof. For nonreal z ,

$$\operatorname{Im} H(z) = (\operatorname{Im} z) \int_0^1 \frac{u d\mu(u)}{|1-zu|^2}.$$

The integral is strictly positive because the measure has positive density except at finitely many interior points. For real $x < 1$, the integrand defining $H(x)$ is positive. Thus H has no zero on Ω . All zeros of D are $1/r_j > 1$, outside Ω , while $D(0) = 1$ and $H(0) = C > 0$. The factorization proves the claimed multiplicity and the global nonvanishing assertion. $\square$

Theorem 4.3 (Minimality and uniqueness of the compensator). *Among real polynomials E with $E(0) > 0$ for which $E(z)L(z)/z^d$ is the transform of a finite positive measure on $[0, 1]$, the smallest possible degree is $d - 1$. Every such polynomial of degree $d - 1$ is $E(0)D(z)$. In particular the normalization $E(0) = 1$ singles out D .*

Proof. Existence in degree $d - 1$ follows from Theorem 4.2. Suppose that $\deg E \leq d - 1$. Lemma 4.1 already realizes the transform as the finite signed measure with density

$$Q_E(u)\kappa(u), \quad Q_E(u) = u^{d-1}E(1/u).$$

Uniqueness of finite Hausdorff moments implies that any purported positive representing measure must be this signed measure. Its density must therefore be nonnegative almost everywhere. Since κ changes sign at every r_j , continuity forces $Q_E(r_j) = 0$. As $r_j \neq 0$, these are $d - 1$ distinct roots $1/r_j$ of E . Consequently $\deg E \geq d - 1$. At equality those roots determine E up to its constant factor, which is fixed by $E(0)$. $\square$

This is minimality of *polynomial compensation for an ordinary positive Stieltjes representation*. It is not minimal depth in an algebra of numerical periods and not a minimal generalized Stieltjes order. Those are different questions.

4.1 Quantitative consequences

For $x > 0$, positivity and $0 < u < 1$ give

$$\frac{Cx^d}{(1+x)D(-x)} < (-1)^d L(-x) < \frac{Cx^d}{D(-x)}. \quad (4.4)$$

Indeed $C/(1+x) < H(-x) < C$ and $D(-x) > 0$. Without computing the compensator roots one still has the coarser bounds

$$C \left(\frac{x}{1+x} \right)^d < (-1)^d L(-x) < Cx^d, \quad (4.5)$$

because $1 \leq D(-x) \leq (1+x)^{d-1}$, with the relevant strictness supplied by H even at depth one.

Write $P(u) = \sum_{j=0}^{d-1} p_j u^j$ and

$$h_n = \int_0^1 u^n d\mu(u) = \sum_{j=0}^{d-1} p_j c_s(n+j+1), \quad n \geq 0. \quad (4.6)$$

For all $n, k \geq 0$,

$$(-1)^k \Delta^k h_n = \int_0^1 u^n (1-u)^k d\mu(u) > 0, \quad \Delta h_n = h_{n+1} - h_n. \quad (4.7)$$

Every finite Hankel matrix $(h_{i+j})_{0 \leq i, j \leq m}$ is positive definite. The same is true of (h_{i+j+1}) and $(h_{i+j} - h_{i+j+1})$. To see this, pair each matrix with a nonzero real vector and integrate the square of its polynomial against, respectively, μ , $u\mu$, or $(1-u)\mu$. Each integral is strictly positive. In particular,

$$h_n h_{n+2} - h_{n+1}^2 > 0.$$

These are exact nonlinear positivity constraints on the original nested harmonic coefficients and the compensator roots. They should not be mistaken for rational-coefficient identities among special values.

5 Exactly $d-1$ angular zeros on every upper semicircle

The signed transform gives, for $0 < \rho \leq 1$ and $0 < \theta < \pi$,

$$\operatorname{Im} L(\rho e^{i\theta}) = \rho \sin \theta J_\rho(\cos \theta), \quad J_\rho(c) = \int_0^1 \frac{\kappa(u)}{1 - 2\rho c u + \rho^2 u^2} du. \quad (5.1)$$

A bound based only on simple sign-change arguments would leave open the possibility of tangencies. We therefore prove a multiplicity-sensitive upper bound before proving existence.

5.1 An elementary confluent Cauchy-kernel bound

Lemma 5.1 (Angular variation bound with multiplicities). *Suppose $\kappa \in L^1(0, 1)$ is continuous, has exactly m sign changes at $r_1 < \dots < r_m$, and has a nonzero fixed sign on each complementary interval. Then J_ρ has at most m zeros in $(-1, 1)$, counted with analytic multiplicity, for every $0 < \rho \leq 1$.*

Proof. The function

$$t(u) = \frac{1 + \rho^2 u^2}{2\rho u} \quad (5.2)$$

is strictly decreasing on $(0, 1)$, since $t'(u) = (\rho^2 u^2 - 1)/(2\rho u^2) < 0$. Its values are at least one, and it tends to infinity as $u \downarrow 0$. Thus

$$J_\rho(c) = \int_0^1 \frac{\kappa(u)}{2\rho u} \frac{1}{t(u) - c} du.$$

The factor $\kappa(u)/u$ need not be integrable by itself; this causes no difficulty because the displayed Cauchy factor is $O(u)$ at zero. All the integrals below are interpreted with their complete kernels.

Suppose there are distinct zeros $c_1, \dots, c_q \in (-1, 1)$ with positive multiplicities $\ell_1, \dots, \ell_q$ summing to $m+1$. If more zeros are available, retain any subcollection of total multiplicity $m+1$. Form the proper rational function

$$R(t) = \frac{\prod_{j=1}^m (t - t(r_j))}{\prod_{a=1}^q (t - c_a)^{\ell_a}}. \quad (5.3)$$

Its partial fractions are linear combinations of $(t - c_a)^{-h}$ for $1 \leq h \leq \ell_a$. Differentiation of J_ρ under the integral is valid locally for $c \in (-1, 1)$; explicitly,

$$J_\rho^{(h-1)}(c) = (h-1)! \int_0^1 \frac{\kappa(u)}{2\rho u} \frac{1}{(t(u) - c)^h} du.$$

The selected zero multiplicities therefore imply

$$\int_0^1 \frac{\kappa(u)}{2\rho u} R(t(u)) du = 0. \quad (5.4)$$

This integral is absolutely convergent: near zero $R(t) = O(1/t) = O(u)$, and near one its denominators stay positive and bounded away from zero.

But each factor $t(u) - t(r_j)$ changes sign exactly where κ changes sign. The denominator of $R(t(u))$ is strictly positive, because $t(u) \geq 1 > c_a$. Hence the integrand in (5.4) has one fixed nonzero sign except at the r_j . Its integral cannot vanish. This contradiction proves the bound and also excludes an identically zero J_ρ . $\square$

The proof is a finite partial-fraction argument, including repeated poles. No unproved strict-total-positivity assertion is being used.

5.2 Positive compensation supplies the missing zeros

For $0 < \theta < \pi$ and $0 \leq u \leq 1$, put

$$\beta_u(\theta) = -\text{Arg}(1 - \rho u e^{i\theta}).$$

For $0 < u < 1$ one has

$$0 < \beta_u(\theta) < \beta_1(\theta) < \frac{\pi}{2}. \quad (5.5)$$

The angle is strictly increasing in u , since differentiation with respect to u gives $\rho \sin \theta / |1 - \rho u e^{i\theta}|^2 > 0$. A positive mixture of the resolvents $(1 - zu)^{-1}$ stays strictly inside this sector. Consequently, with H as in (4.3),

$$0 < \text{Arg } H(\rho e^{i\theta}) < \beta_1(\theta). \quad (5.6)$$

For example, the second inequality follows by rotating each resolvent through $-\beta_1$ and observing that its imaginary part is strictly negative for $0 < u < 1$. All rays lie in an angle of width less than $\pi/2$, so no argument ambiguity arises.

Theorem 5.2 (Complete angular count and signs). *For (1.2) and every $0 < \rho \leq 1$, $\text{Im } L(\rho e^{i\theta})$ has exactly $d - 1$ simple zeros in $0 < \theta < \pi$. They have the signs and bounds stated in Theorem 1.1.*

Proof. A continuous unwrapped argument of L along the upper arc is

$$\Phi(\theta) = d\theta + \sum_{j=1}^{d-1} \beta_{r_j}(\theta) + \text{Arg } H(\rho e^{i\theta}). \quad (5.7)$$

The factorization proves this formula exactly; Φ is not reduced modulo 2π . The sector bounds imply

$$d\theta < \Phi(\theta) < d(\theta + \beta_1(\theta)) < d\pi. \quad (5.8)$$

At the other end of the arc, $\Phi(\theta) \rightarrow d\pi$ as $\theta \uparrow \pi$, since all resolvents approach positive real values and $H(-\rho) > 0$.

If $\rho < 1$, then $\Phi(\theta) \rightarrow 0$ as $\theta \downarrow 0$. The case $\rho = 1$ requires more care: $L(1)$ may be singular, and there is no reason to assign it argument zero. Instead, each fixed $r_j < 1$ gives $\beta_{r_j}(\theta) \rightarrow 0$, while (5.6) gives

$$\limsup_{\theta \downarrow 0} \text{Arg } H(e^{i\theta}) \leq \pi/2.$$

It follows that $0 < \Phi(\theta) < \pi$ for sufficiently small positive θ . This weaker endpoint statement is all that is needed.

By continuity, the phase crosses each of $\pi, 2\pi, \dots, (d-1)\pi$. These yield at least $d-1$ distinct zeros of the imaginary part. On the other hand, Lemma 5.1 with $m = d-1$, together with (5.1), gives at most $d-1$ zeros counted with multiplicities. The change of variable $c = \cos \theta$ has nonzero derivative in the open arc, and the prefactor $\rho \sin \theta$ is positive. Thus multiplicities are preserved. Every zero is simple, and there is exactly one crossing of each listed phase level. The levels must occur in their stated order, since a reversal would force a repeated crossing of an earlier level.

The initial phase lies in $(0, \pi)$, so the imaginary part starts positive. Simplicity forces alternating signs thereafter. At the k th zero, the phase is $k\pi$ and the modulus is nonzero, so the real value has sign $(-1)^k$.

For the location bounds, write $\alpha_k = k\pi/d$. At a zero, (5.8) gives

$$\theta_k < \alpha_k < \theta_k + \beta_1(\theta_k).$$

The function $h(\theta) = \theta + \beta_1(\theta)$ is strictly increasing, because

$$h'(\theta) = \frac{1 - \rho \cos \theta}{1 - 2\rho \cos \theta + \rho^2} > 0.$$

For $\rho < 1$, solving $h(\theta) = \alpha$ gives $\theta = \alpha - \arcsin(\rho \sin \alpha)$. At $\rho = 1$ the same expression is the solution when $\alpha > \pi/2$; when $\alpha \leq \pi/2$ it is zero, and the strict lower bound follows from $\theta_k > 0$. This proves (1.5). $\square$

The argument proves the needed crossings without asserting that Φ is globally monotone. Such an assertion is unnecessary here.

Corollary 5.3 (Analytic motion and no interior collisions). *For fixed integer leading indices, each angular branch is real analytic in $(\rho, b) \in (0, 1) \times (0, \infty)$, and extends real analytically across $\rho = 1$ locally at each of its nontrivial upper-arc zeros. The density roots r_j are real analytic in $b > 0$. No two of the angular branches collide for $0 < \rho \leq 1$.*

Proof. On compact subsets of $\text{Re } b > 0$, the seed f_b is holomorphic as an L^1 -valued function; factors $|\log(-\log u)|^j$ arising from parameter differentiation have an integrable common majorant on smaller compact subsets. The bounded operators preserve this holomorphic parameter dependence. The signed transform is locally holomorphic in $z \in \Omega$ and in b . Simplicity of its angular zeros permits the real analytic implicit-function theorem. The unit-radius zeros remain away from the cut, so the same local argument crosses $\rho = 1$. The density derivative formulas give joint real analyticity in u, b in the open interval, and their simple roots obey the same theorem. Distinctness follows from Theorem 5.2. $\square$

The local extension across radius one does not imply that the global count persists at all larger radii; Section 9 gives an explicit obstruction.

5.3 Strict radial motion of every angular branch

The compensation yields more than a zero count. It also turns a monotonicity property of ordinary positive Stieltjes transforms into strict motion of all the strict-depth angular zeros.

Lemma 5.4 (Radial argument monotonicity for positive transforms). *Let ν be a nonzero finite positive measure on $[0, 1]$ and $Q(z) = \int (1 - uz)^{-1} d\nu(u)$. For $0 < \theta < \pi$ and $0 < \rho \leq 1$,*

$$\frac{\partial}{\partial \rho} \text{Arg } Q(\rho e^{i\theta}) \geq 0. \quad (5.9)$$

The argument is chosen continuously from $Q(0) > 0$.

Proof. First take a finite measure with positive weights at distinct points $0 < u_1 < \dots < u_n < 1$. The rational function $\sum_j w_j/(t - u_j)$ is strictly decreasing between consecutive poles, with limits $+\infty$ and $-\infty$ at the two ends. Its numerator therefore has exactly one root $v_j \in (u_j, u_{j+1})$ in each such interval. Factoring the numerator gives

$$Q(z) = Q(0) \frac{\prod_{j=1}^{n-1} (1 - v_j z)}{\prod_{j=1}^n (1 - u_j z)}.$$

Consequently the radial derivative of its argument is

$$\sin \theta \left\{ f(u_1) + \sum_{j=1}^{n-1} (f(u_{j+1}) - f(v_j)) \right\}, \quad f(t) = \frac{t}{1 - 2\rho t \cos \theta + \rho^2 t^2}.$$

Every summand is nonnegative, because

$$f'(t) = \frac{1 - \rho^2 t^2}{(1 - 2\rho t \cos \theta + \rho^2 t^2)^2} \geq 0 \quad (0 \leq t \leq 1).$$

Approximate an arbitrary finite positive ν weakly by such finite measures, with their masses converging to $\nu([0, 1])$. For each compact subset of Ω , the integrands defining the transform and its first complex derivative are bounded and continuous uniformly in $u \in [0, 1]$. The transforms and derivatives converge locally uniformly. Also $Q(\rho e^{i\theta}) \neq 0$: it has positive imaginary part unless the measure is supported at zero, in which case it is a positive constant. Passing to the limit in $\text{Im}(e^{i\theta} Q'(z)/Q(z))$ proves the nonnegative derivative. The same argument works at $\rho = 1$ because θ is strictly between zero and π . $\square$

Theorem 5.5 (Strict inward angular motion). *For $d \geq 2$ and (1.2), every angular branch satisfies*

$$\frac{d\theta_k}{d\rho} < 0, \quad 0 < \rho \leq 1, \quad 1 \leq k \leq d-1. \quad (5.10)$$

The derivative at one is the one-sided derivative, or equivalently the derivative of the local continuation. The same result holds for the positive-seed and shifted-terminal families of Theorem 3.4.

Proof. At fixed θ , differentiate the unwrapped phase (5.7) with respect to the radius:

$$\Phi_\rho = \sum_{j=1}^{d-1} \frac{r_j \sin \theta}{1 - 2\rho r_j \cos \theta + \rho^2 r_j^2} + \partial_\rho \text{Arg } H(\rho e^{i\theta}) > 0.$$

The first sum is strictly positive since $d \geq 2$ and $r_j > 0$; the last term is nonnegative by Lemma 5.4. At the unique simple crossing of the phase level $k\pi$, one has $\Phi_\theta > 0$. Indeed the phase starts below that level, ends above it, and crosses it exactly once; simplicity excludes a zero phase derivative there. Implicit differentiation of $\Phi(\theta_k(\rho), \rho) = k\pi$ now gives

$$\theta'_k(\rho) = -\frac{\Phi_\rho}{\Phi_\theta} < 0.$$

All ingredients also hold for a general positive seed. $\square$

For depth two, this does *not* settle the manuscript's conjecture about $\eta(\rho) = \cos \theta(\rho)/\rho$: the derivative of that quotient has a second term, and its sign is not implied by $\theta'(\rho) < 0$. The two kinds of radial monotonicity must remain separate in the integration ledger.

6 Small-radius and large-outer-index expansions

The global count removes an ambiguity from local expansions: the zeros found near the first Fourier-mode zeros are the entire list of angular zeros, not just selected branches.

Theorem 6.1 (Two-term radial expansion). *Fix an index satisfying (1.2) with $d \geq 2$, and put*

$$p = \frac{c_s(d+1)}{C_s}, \quad q = \frac{c_s(d+2)}{C_s}, \quad \alpha_k = \frac{k\pi}{d}.$$

The k th angular branch extends real analytically to $\rho = 0$, and

$$\begin{aligned}\theta_k(\rho) &= \alpha_k - \frac{p}{d} \sin \alpha_k \rho \\ &\quad + \left(\frac{d+1}{d^2} p^2 - \frac{2q}{d} \right) \sin \alpha_k \cos \alpha_k \rho^2 + O_s(\rho^3).\end{aligned}\tag{6.1}$$

In particular, every branch initially moves strictly toward smaller angles as the radius increases from zero.

Proof. Divide the imaginary part of the disk series by $C_s \rho^d$. The resulting analytic equation is

$$0 = \sin(d\theta) + p\rho \sin((d+1)\theta) + q\rho^2 \sin((d+2)\theta) + O(\rho^3).$$

At $(\rho, \theta) = (0, \alpha_k)$ its θ derivative is $d(-1)^k \neq 0$, so the analytic implicit-function theorem applies. There are $d-1$ such branches; for small positive radius they exhaust the global list from Theorem 5.2.

Write $\theta = \alpha_k + A\rho + B\rho^2 + O(\rho^3)$ and divide by $(-1)^k$. The coefficients of ρ and ρ^2 are respectively

$$dA + p \sin \alpha_k, \quad dB + (d+1)pA \cos \alpha_k + q \sin(2\alpha_k).$$

Solving gives (6.1). The first derivative is negative because $p > 0$ and $0 < \alpha_k < \pi$. $\square$

Higher Taylor coefficients are obtained by the same finite coefficient recursion. At each order the new unknown occurs with coefficient $d(-1)^k$, while all remaining terms depend only on earlier coefficients and finitely many moments. This is an exact algebraic recurrence, not a proof of any global radial sign.

6.1 Uniformity as the first index increases

Fix $d \geq 2$ and positive integer middle indices $s_2, \dots, s_{d-1}$. Let the first index be $a \in \mathbb{N}$ and the last be $b > 0$. Define

$$T_N(b) = \sum_{N \geq n_2 > \dots > n_d \geq 1} \frac{1}{n_2^{s_2} \dots n_{d-1}^{s_{d-1}} n_d^b}, \quad T_* = T_{d-1}(b) > 0.$$

The latter number is a single-chain value and is independent of b . At depth two, $T_N(b) = H_N^{(b)}$ and $T_* = 1$.

Theorem 6.2 (First displacement, uniformly in the final order). *Set $\lambda = d/(d+1)$ and $\mu = d/(d+2)$. As $a \rightarrow \infty$ through positive integers, uniformly for $b > 0$, $0 \leq \rho \leq 1$, and $1 \leq k \leq d-1$,*

$$\theta_k(\rho) = \alpha_k - \frac{T_d(b)}{dT_*} \sin \alpha_k \rho \lambda^a + O(\rho^2 \mu^a).\tag{6.2}$$

At $\rho = 0$, the branch is interpreted by its analytic extension. The implied constant and the starting threshold for a may depend on the fixed depth and middle indices, but not on b , ρ , or k .

Proof. The normalized imaginary part is

$$\sin(d\theta) + \frac{T_d(b)}{T_*} \rho \lambda^a \sin((d+1)\theta) + R(\theta),\tag{6.3}$$

where

$$R(\theta) = \sum_{n \geq d+2} \frac{T_{n-1}(b)}{T_*} \rho^{n-d} \left(\frac{d}{n} \right)^a \sin(n\theta).$$

All inner denominators are at least one, so $0 \leq T_{n-1}(b) \leq n^{d-1}$ uniformly for $b > 0$. Choose the fixed integer $a_0 = d+3$. For $a \geq a_0$, termwise differentiation is absolutely justified and

$$|R(\theta)| + |R'(\theta)| \leq K \rho^2 \mu^a, \quad K = \frac{2}{T_*} \sum_{n \geq d+2} n^d \left(\frac{d+2}{n} \right)^{a_0} < \infty.\tag{6.4}$$

Also $T_d(b)/T_* \leq d^{d-1}/T_*$, independently of b .

On disjoint fixed small neighborhoods of the α_k , the derivative of the first term of (6.3) is bounded away from zero. Its perturbation tends to zero uniformly by (6.4). Evaluating at $\alpha_k \pm K_1 \rho \lambda^a$, for a sufficiently large fixed K_1 , gives opposite signs for all sufficiently large a . This follows from the linear term $d(\theta - \alpha_k)$ and the uniform bounds on the perturbations; $\mu/\lambda < 1$. Thus each neighborhood contains a unique root with displacement $\delta = O(\rho \lambda^a)$. The global count identifies these roots as all the branches in their established order.

Expand (6.3) at α_k and divide by $(-1)^k$. The root equation becomes

$$0 = d\delta + \frac{T_d(b)}{T_*} \rho \lambda^a \sin \alpha_k + O(\delta^2 + \rho \lambda^a |\delta| + \rho^2 \mu^a).$$

Because $\lambda^2 < \mu$, the previously obtained displacement bound makes the error $O(\rho^2 \mu^a)$. Division by d proves (6.2). All constants used above are independent of the final order. $\square$

At depth two and unit radius this recovers the leading scale $(2/3)^a$ already present in the manuscript's more detailed depth-two expansion [1]. The added content is the simultaneous all-depth statement, the exact branch indexing, and uniformity over the entire range $b > 0$ and $0 \leq \rho \leq 1$.

7 Explicit logistic kernels and positive integral identities

The compensator is generally implicit: its roots are the zeros of a signed density. The all-ones family makes every part explicit and gives an infinite collection of exact positive integral identities.

7.1 The all-ones density and its roots

Let $\mathbf{1}_d = (1, \dots, 1)$ and define

$$\ell(u) = \log \frac{u}{1-u}, \quad 0 < u < 1.$$

The familiar identity

$$L_{\mathbf{1}_d}(z) = \frac{(-\log(1-z))^d}{d!} \tag{7.1}$$

follows recursively from $(1-z)L'_{\mathbf{1}_d}(z) = L_{\mathbf{1}_{d-1}}(z)$ and the vanishing initial value at zero. We use it as a classical identity with a short proof, not as a new reduction of multiple polylogarithms.

Theorem 7.1 (Explicit signed kernels and logistic roots). *For every $d \geq 1$,*

$$\begin{aligned} \kappa_{\mathbf{1}_d}(u) &= \frac{\operatorname{Im}(\ell(u) + i\pi)^d}{\pi d!} \\ &= \sum_{j=0}^{\lfloor (d-1)/2 \rfloor} \frac{(-1)^j \pi^{2j} \ell(u)^{d-1-2j}}{(2j+1)!(d-1-2j)!}. \end{aligned} \tag{7.2}$$

Its increasing list of roots is

$$r_j = \frac{1}{1 + \exp(\pi \cot(j\pi/d))}, \quad 1 \leq j \leq d-1. \tag{7.3}$$

In particular, $r_j + r_{d-j} = 1$.

Proof. Consider the generating function

$$G(t, u) = e^{t\ell(u)} \frac{\sin \pi t}{\pi t} = \sum_{d \geq 1} K_d(u) t^{d-1}. \tag{7.4}$$

For $|\operatorname{Re} t| < 1$, its integral at moment one is the beta integral

$$\int_0^1 G(t, u) \, du = \frac{\sin \pi t}{\pi t} \Gamma(1+t) \Gamma(1-t) = 1.$$

The reflection formula supplies the last equality, with removable values understood at zero. Thus $\int K_1 = 1$ and $\int K_d = 0$ for $d \geq 2$. Coefficient extraction is legitimate on a smaller complex neighborhood of zero, since $u^{\operatorname{Re} t} (1-u)^{-\operatorname{Re} t}$ and its logarithmic parameter derivatives have an integrable majorant.

Now $K_1 = 1$ and

$$\frac{d}{du} G(t, u) = \frac{t}{u(1-u)} G(t, u),$$

so $K'_d = K_{d-1}/(u(1-u))$. This is exactly the derivative of BK_{d-1} . Their difference is constant, and both have zero integral for $d \geq 2$, so $K_d = BK_{d-1}$. Hence these are the constructed all-ones kernels. Expanding the exponential and sine gives (7.2).

The imaginary part of $(\ell + i\pi)^d$ vanishes when $\operatorname{Arg}(\ell + i\pi)$ is one of $j\pi/d$. This gives $\ell = \pi \cot(j\pi/d)$. Reversing the order of these real values and solving $u/(1-u) = e^\ell$ yields (7.3). The symmetry follows from $\cot(\pi - x) = -\cot x$. $\square$

Proposition 7.2 (A continuous Mellin identity and Stirling moments). *For real $x > 0$ and complex t satisfying $-x < \operatorname{Re} t < 1$,*

$$\int_0^1 u^{x-1} e^{t\ell(u)} \frac{\sin \pi t}{\pi t} \, du = \frac{\Gamma(x+t)}{\Gamma(x+1)\Gamma(1+t)}. \quad (7.5)$$

The values at removable singularities are obtained by continuation. For every integer $n \geq 1$,

$$\int_0^1 u^{n-1} \kappa_{\mathbf{1}_d}(u) \, du = \frac{\left[\begin{smallmatrix} n \\ d \end{smallmatrix} \right]}{n!}, \quad (7.6)$$

where $\left[\begin{smallmatrix} n \\ d \end{smallmatrix} \right]$ is the unsigned Stirling number of the first kind, zero for $d > n$.

Proof. The integral equals $\sin(\pi t) B(x+t, 1-t)/(\pi t)$. The beta formula and Gamma reflection simplify it to (7.5). At integer $x = n$, the right side is

$$\frac{(t+1)(t+2) \cdots (t+n-1)}{n!}.$$

Its coefficient of t^{d-1} is $\left[\begin{smallmatrix} n \\ d \end{smallmatrix} \right]/n!$, proving (7.6) by (7.4). $\square$

These beta and Stirling identities provide an independent normalization check on the general signed-moment construction.

7.2 An explicit infinite family of compensated identities

For the roots (7.3), define

$$P_d(u) = \prod_{j=1}^{d-1} (u - r_j), \quad D_d(z) = \prod_{j=1}^{d-1} (1 - r_j z).$$

The integrable function $P_d \kappa_{\mathbf{1}_d}$ is strictly positive away from its finitely many zeros. Combining (4.3) and (7.1) gives the exact identity

$$\boxed{\int_0^1 \frac{P_d(u) \kappa_{\mathbf{1}_d}(u)}{1-zu} \, du = \frac{D_d(z)}{d!} \left(\frac{-\log(1-z)}{z} \right)^d, \quad z \in \Omega.} \quad (7.7)$$

At $z = 0$ the removable value is $1/d!$. At $z = -1$ this becomes

$$\int_0^1 \frac{P_d(u)\kappa_{1_d}(u)}{1+u} du = \frac{D_d(-1)}{d!} \log^d 2. \quad (7.8)$$

The content is not a new formula for L_{1_d} ; it is the explicit positive density, its canonical compensation, and the resulting positive-integrand identities. They remain valid at every depth without a numerical relation search.

7.3 Depth three

Put

$$h_3 = \frac{\pi}{\sqrt{3}}, \quad q_3 = \frac{1}{2 + 2 \cosh h_3}.$$

Then

$$P_3(u) = u^2 - u + q_3, \quad D_3(z) = 1 - z + q_3 z^2, \quad \kappa_{1_3}(u) = \frac{\ell(u)^2 - \pi^2/3}{2}.$$

Both factors P_3 and $\ell^2 - \pi^2/3$ are negative between the same two roots and positive outside them. Hence their product is nonnegative throughout the interval. Formula (7.8) gives

$$\boxed{\int_0^1 \frac{(u^2 - u + q_3)(\ell(u)^2 - \pi^2/3)}{1+u} du = \frac{2+q_3}{3} \log^3 2.} \quad (7.9)$$

The companion normalization is

$$\int_0^1 (u^2 - u + q_3)(\ell(u)^2 - \pi^2/3) du = \frac{1}{3}. \quad (7.10)$$

These identities are proved statements, not high-precision conjectures.

7.4 Depth four

Set

$$q_4 = \frac{1}{2 + 2 \cosh \pi}, \quad P_4(u) = \left(u - \frac{1}{2}\right)(u^2 - u + q_4).$$

Now

$$D_4(z) = \left(1 - \frac{z}{2}\right)(1 - z + q_4 z^2), \quad \kappa_{1_4}(u) = \frac{\ell(u)^3 - \pi^2 \ell(u)}{6}.$$

The roots are the logistic images of $-\pi, 0, \pi$, so again the product $P_4 \kappa_{1_4}$ is nonnegative. The corresponding pair of identities is

$$\int_0^1 \frac{P_4(u)(\ell(u)^3 - \pi^2 \ell(u))}{1+u} du = \frac{3(2+q_4)}{8} \log^4 2, \quad (7.11)$$

$$\int_0^1 P_4(u)(\ell(u)^3 - \pi^2 \ell(u)) du = \frac{1}{4}. \quad (7.12)$$

All endpoint singularities are logarithmic powers and are integrable. Thus no principal-value convention is implicit in these formulas.

7.5 An explicit density-edge asymptotic

The smallest all-ones density root gives a simple large-depth benchmark:

$$\log r_1 = -d + \frac{\pi^2}{3d} + \frac{\pi^4}{45d^3} + O(d^{-5}) + O(e^{-d}). \quad (7.13)$$

Indeed $\log r_1 = -\pi \cot(\pi/d) - \log(1 + e^{-\pi \cot(\pi/d)})$, and the Taylor expansion of $\cot$ at zero gives the displayed terms. This concerns a density root, not an angular zero. Symmetry supplies $1 - r_{d-1} = r_1$.

8 Exact rational evaluation and finite certificates

The density zeros need not be computed in order to obtain numerical certificates. The unsigned norm bound $\|\kappa\|_1 \leq 2^{d-1}$ already gives an explicit geometric remainder for a centered resolvent expansion. The proof works with real final order; the delivered exact implementation accepts positive integer indices so that all required moments are rational.

8.1 A centered-moment expansion

Define

$$B_k = \int_0^1 (2u-1)^k \kappa(u) du = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} 2^j c_s(j+1). \quad (8.1)$$

Then $|B_k| \leq M$, where $M = 2^{d-1}$. For $\operatorname{Re} z < 1$, put

$$w = \frac{z}{2-z}, \quad |w| < 1.$$

The inequality is equivalent to $|z| < |2-z|$. The identity $1-zu = (1-z/2)(1-w(2u-1))$ gives

Theorem 8.1 (Geometric exact-evaluation certificate). *For every integer $N \geq 1$,*

$$V_N(z) = \frac{z}{1-z/2} \sum_{k=0}^{N-1} B_k w^k, \quad (8.2)$$

$$|L(z) - V_N(z)| \leq \left| \frac{z}{1-z/2} \right| \frac{M|w|^N}{1-|w|}. \quad (8.3)$$

At $z = i$ and even $N = 2m$, a purely rational bound is

$$|L(i) - V_{2m}(i)| < \frac{2^d}{5^m}. \quad (8.4)$$

For $|z| \leq 1/2$, the uniform bound is

$$|L(z) - V_N(z)| \leq \frac{2^{d-1}}{3^N}. \quad (8.5)$$

Proof. The centered geometric series converges uniformly for $u \in [0, 1]$, and its tail is bounded by $|w|^N/(1-|w|)$. Integrating against $|\kappa|$ proves (8.3). At $z = i$, $|w| = 1/\sqrt{5}$ and $|z/(1-z/2)| = 2/\sqrt{5}$. The multiplicative constant is therefore $2M/(\sqrt{5}-1) < 2M = 2^d$ after extracting 5^{-m} . On $|z| \leq 1/2$, one has $|w| \leq 1/3$ and $|z/(1-z/2)| \leq 2/3$, proving (8.5). $\square$

For integer indices, (8.1) and (8.2) use only rational arithmetic at rational complex z . An enclosing complex disk gives enclosing intervals for each coordinate by adding and subtracting its radius. The evaluator never treats a small residual as a proof of an identity.

8.2 Implementation choices and replay

The file `code/exact_evaluator.py` first obtains the moments by dynamic strict prefix sums. It computes B_k with a common integer denominator and a repeated-difference transform, rather than a floating-point alternating binomial sum. Horner evaluation then gives the exact rational center. The generic branch can upper-bound square roots by dyadic rationals; the Gaussian and half-disk cases use the closed rational bounds above. Near $\operatorname{Re} z = 1$, the fixed norm-bound precision in the generic branch can be insufficient, in which case the implementation raises an error rather than returns an uncertified value. It is not advertised as a complete optimal evaluator on the entire slit plane.

From the package root, the principal replay is

`python code/verify.py`

It reads frozen rational certificates and independently recomputes them. The optional command

`python code/verify.py --regenerate`

uses floating point to *propose* angular brackets, then requires rational arithmetic to certify both endpoint signs before saving them. The proposal stage uses a strict-disk series and does not decide any accepted sign. This discovery/replay separation is deliberate.

8.3 Gaussian values

The delivered Gaussian certificate file contains eight mixed-depth examples, all evaluated with $N = 240$ centered terms. Each complex-disk radius is smaller than 10^{-81} . The decimal entries below are readable midpoint diagnostics; the certificates are the rational endpoints in `data/gaussian_enclosures.json`.

Index	Re $L_s(i)$, approximately	Im $L_s(i)$, approximately
(2, 1)	-0.174715661158	-0.113648917600
(1, 1, 1)	0.099953993665	-0.033577147847
(2, 1, 2)	0.032383901801	-0.026348629072
(1, 2, 1, 1)	0.001158016810	0.008343470492
(2, 1, 1, 2)	0.001825088603	0.006510815484
(1, 1, 1, 1, 1)	-0.003396506582	-0.001886779708
(2, 1, 2, 1, 1)	-0.000326599679	-0.000010475684
(1, 1, 1, 1, 1, 1)	0.000443168800	-0.000335617002

In addition, the eight all-ones Gaussian values at depths one through eight are checked against *independent exact* intervals for

$$L_{\mathbf{1}_d}(i) = \frac{(-\frac{1}{2} \log 2 + \frac{i\pi}{4})^d}{d!}.$$

The logarithm and π intervals are derived from different rational series, as detailed in Appendix A. These eight normalization controls are distinct from the eight stored mixed-depth examples above.

8.4 Certified angular brackets

On the semicircle of radius $1/2$, set

$$z(t) = \frac{1-t^2}{2(1+t^2)} + i \frac{t}{1+t^2}, \quad \theta = 2 \arctan t, \quad t > 0. \quad (8.6)$$

Rational t gives rational real and imaginary coordinates. Every saved root bracket has t -width at most 10^{-12} ; the corresponding angle-width is less than $2 \cdot 10^{-12}$. With $N = 104$, the error bound (8.5) proves the two endpoint signs by strict rational comparisons. For illustration:

Index	k	$10^{12}t_{\text{left}}$	$10^{12}t_{\text{right}}$
(2, 1, 2)	1	472099547153	472099547154
(2, 1, 2)	2	1487963538987	1487963538988
(1, 2, 1, 1)	1	314972569528	314972569529
(1, 2, 1, 1)	2	806524527583	806524527584
(1, 2, 1, 1)	3	2019771345242	2019771345243
(1, 1, 1, 1, 1, 1)	3	774596669241	774596669242

The file `data/angular_brackets.json` contains 24 brackets in eight complete index families of depths two through six, together with all exact coordinate intervals. The existence of a zero in each bracket follows from the opposite signs. The theorem, rather than a scan, proves uniqueness in every bracket and excludes all additional angular zeros. For the displayed middle branch at depth six, the all-ones formula independently gives $\theta_3(1/2) = \arccos(1/4)$ and $t = \sqrt{3/5}$.

8.5 What the exact suite checks

The fresh replay reports 5,713 finite check groups:

Check family	Count
Strict coefficients against direct finite nested enumeration	1080
Initial zero moments and leading coefficients	120
Centered moments: independent binomial formula and norm bound	2880
Polynomial compensation identities for arbitrary test roots	1200
All-ones moments against unsigned Stirling recurrence	240
Logistic-kernel derivatives through depth twelve	11
Depth-three/four symbolic kernel formulas	2
Exact generalized-kernel counterexample checks	2
Rejected negative controls	2
Rational atomic radial-argument derivative signs	80
Independent exact all-ones Gaussian controls	8
Replayed stored Gaussian enclosures	8
Replayed angular endpoint signs	48
Replayed root brackets and complete families	24 + 8
Total	5713

The polynomial compensation tests deliberately use arbitrary rational roots: they test the cancellation identity, *not* positivity for those roots. Positivity requires the actual density zeros and is proved analytically. Similarly, finite atomic derivative checks do not replace the interlacing-and-approximation proof of Lemma 5.4.

9 Audit findings, corrections, and failed shortcuts

The source review was targeted, not a line-by-line audit of the entire canonical manuscript. It covered the signed-kernel, real-order, positive-double-kernel, subcritical, local-radius, and research-status sections. No substantive counterexample to the reviewed depth-two kernel and zero theorems was found. The following two editorial corrections are concrete; the later items are scope safeguards for this continuation rather than allegations that the manuscript makes those mistakes.

9.1 Two notation corrections in the transport proof

In `chapters/05-real-positive-kernel.tex`, at the pinned snapshot, the proof of the Hurwitz–polylogarithm transport formula says to specialize the regularized Hurwitz identity “with $q = n$ ”, and then invokes boundedness of “ h ” [4]. The displayed Hurwitz formula uses x as its real argument, and the regularized kernel is named $q(t)$. The intended text is therefore

$$\text{“with } x = n\text{”,} \quad \text{“boundedness of } q\text{”}.$$

The theorem and the coefficient calculation are unchanged. The file `integration/notation_corrections.patch` contains precisely these two replacements, at source lines 102 and 104. It is an optional proposed patch; it has not been applied to the remote repository.

9.2 Positive higher-order resolvents need not be zero-free

The following exact example shows why the oscillation step in this article cannot be replaced by a generic positivity slogan:

$$R(z) = z^4 \left(1 + \frac{1}{(1-z)^4} \right) = z^4 \int_{[0,1]} \frac{d(\delta_0 + \delta_1)(u)}{(1-uz)^4}. \quad (9.1)$$

The measure is positive, yet

$$z_0 = 1 - e^{-i\pi/4} = 1 - \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$$

is a nonzero zero of R and $|z_0|^2 = 2 - \sqrt{2} < 1$. Indeed $(1 - z_0)^4 = -1$. Thus a positive generalized Stieltjes kernel of order four does not imply even disk nonvanishing. Our ordinary positive transform appears only *after* the sign-change polynomial has been identified and the initial moments have been canceled.

9.3 The unit-radius endpoint cannot be assigned an arbitrary phase

At $\rho = 1$, the point $z = 1$ lies on the removed cut. A proof of angular existence that simply sets $\text{Arg } L(1) = 0$ may silently assume a finite endpoint value or a particular asymptotic. The proof in Section 5 instead uses the sector bound $\text{Arg } H < \beta_1$ to show that the unwrapped phase is below π near the initial endpoint. This suffices even when L diverges there. No theorem about a limiting singular value is smuggled into the argument.

9.4 The angular count does not extend automatically beyond radius one

For the elementary depth-two example $L_{1,1}(z) = \frac{1}{2} \log^2(1-z)$, a zero of its imaginary part in the upper half-plane requires

$$|1 - z| = 1.$$

On $|z| = \rho$, this is $\cos \theta = \rho/2$. For $\rho > 2$ there is no such open-upper-arc zero at all. In our proof, the map $t(u) = (1 + \rho^2 u^2)/(2\rho u)$ ceases to be monotone on $(0, 1)$ when $\rho > 1$, and the derivative used in the radial argument lemma can also change sign. These are genuine losses of hypotheses. Local continuation of a simple zero through $\rho = 1$ must not be confused with a theorem for all radii.

9.5 Integer leading orders, normalized radii, and arithmetic depth

The operator A^{s_j-1} was iterated an integer number of times. Its elementary Rolle argument does not automatically apply to an arbitrary fractional power. The manuscript itself proves that finite signed moment representations can fail in part of the positive real-order depth-two range $[2, 3]$. Therefore one cannot state the present L^1 construction for all positive real indices without a separate theorem or a different representation.

Likewise, strict decrease of $\theta_k(\rho)$ does not imply a chosen sign for $\cos \theta_k(\rho)/\rho$. The original normalized-radius conjecture and the fractional turning-point issues remain separate. Finally, the degree $d - 1$ in the compensation theorem is not a claim about minimal multiple-polylogarithm depth at a numerical argument. Formal relation modules, arithmetic period independence, and positive transform compensation are different mathematical objects.

9.6 The remaining arithmetic candidates retain their status

The reviewed README and discovery chapter distinguish the proved S_4 identity, the unresolved S_6 and new S_8 candidates, and an older S_8 vector that was rigorously rejected [5]. The present analytic theorem does not furnish a relation certificate for S_6 or S_8 . Its Gaussian enclosures also do not establish arithmetic independence or any conjectural equality merely by small residuals. These status labels should remain unchanged when the package is integrated.

10 Further research questions and proposed directions

The proofs above suggest a more structured programme than an unrestricted search for small numerical relations. The following questions separate analytic geometry, constructive certificates, and arithmetic reductions.

10.1 Fractional leading indices

Determine the largest real-index domain on which a finite signed moment density exists and has exactly $d - 1$ sign changes. The first focused case is depth three with a real first index and integer second index. The all-positive depth-two results show that absence of a finite signed measure need not imply failure of zero-freeness. Consequently there are two different problems: extending the present density theorem, and extending the zero theorem by a compensated or distributional method.

For real leading indices at least one, a plausible testable target is that the same $d - 1$ oscillation statement persists. This is a proposed conjectural direction, not a claim supported here by an exhaustive fractional-parameter experiment. A proof would need a substitute for the derivative-zero allocation used by A.

10.2 Normalized angular motion and its higher-depth analogue

Theorem 5.5 settles strict decrease of the angles, but not the manuscript's finer quotient $\eta(\rho) = \cos \theta(\rho)/\rho$. At depth two, retain the existing integer-outer-index normalized-radius conjecture exactly as stated in the manuscript. At higher depth, a more natural displacement is $(\alpha_k - \theta_k(\rho))/\rho$. Determine its monotonicity regions, transition loci, and whether compensator-root interlacing constrains them. The explicit coefficient in (6.1) provides an exact local diagnostic, not a global conclusion.

10.3 Motion under changes of an index or the terminal seed

Raising the first index gives strict interlacing of density zeros. Does it also order every angular zero in the expected direction toward α_k ? Formula (6.2) proves the limiting behavior but does not compare every pair of finite consecutive outer indices. A positive answer would require comparing both the compensator and its positive transform, not only the locations of the compensator roots. Analogous questions for an inner index, the terminal shift λ , and positive terminal mixtures are also well posed.

10.4 Sharper variation norms and faster certificates

The bound $\|\kappa\|_1 \leq 2^{d-1}$ is uniform and easy to certify, but it is not asserted optimal. Determine the best constant at fixed depth, or at fixed depth and first index. The exact logistic kernels supply a useful family in which the variation can be expressed through one-dimensional integrals between known roots. A smaller certified variation bound would directly tighten (8.3). Another target is a full-slit evaluator with comparable exact certificates and a conformal or Chebyshev expansion adapted to the cut.

10.5 Constructive compensator roots and positive approximation

Develop rational interval enclosures for all r_j without assuming closed forms. The proved interlacing laws can supply nested brackets when an index or the depth changes. Once such enclosures are available, combine the positive moment matrices of (4.6) with quadrature or Padé-type approximants to produce two-sided bounds on real arguments. The separation between an exact root theorem and a finite root-isolation certificate should be retained throughout.

10.6 Large-depth geometry

For the all-ones family, (7.3) and (7.13) determine the density-edge scale. For general index patterns, determine whether the edge roots still have exponential scale in depth, and how the leading exponents depend on the index profile. Compare this with the angular boundary layers near $k = 1$ and $k = d - 1$. Uniform asymptotics with both d and the outer index growing would require estimates beyond the fixed-depth remainder in Theorem 6.2.

10.7 Radii beyond one and zero-count transitions

The first target beyond the proved radius range is the explicit all-ones family. There the problem reduces to the argument of $-\log(1 - \rho e^{i\theta})$. Classify the radii at which angular zeros enter or leave through the cut endpoint, or form tangencies. The elementary depth-two loss at $\rho = 2$ shows that a universal continuation without transitions is impossible. The higher-depth problem may distinguish endpoint transitions from interior folds.

10.8 Colored arguments and arithmetic identities

For nontrivial inner colors, the real positive seed or the real signed kernel can disappear. Identify color patterns admitting a real character projection with controlled sign variation. This would be a meaningful route toward analogous zero theorems for cyclotomic multiple polylogarithms. Separately, pursue the manuscript's frozen S_6 and S_8 candidates through exact functional-equation certificates. The all-depth evaluator can supply checks, but a numerical enclosure containing zero is not the missing arithmetic proof.

10.9 A finite-to-analytic formalization path

The first formal targets are the two operator moment identities, finite polynomial cancellation, rational partial fractions with repeated poles, and the exact evaluator's remainder arithmetic. The analytic layer can then isolate Rolle allocation, Hausdorff-measure uniqueness, positive-transform sectors, and the implicit-function argument. None of the current code is represented as a Lean or other proof-assistant formalization. A future formalization should record exactly where classical compactness and analytic continuation enter, rather than labeling the numerical replay as a substitute.

A Independent rational logarithm and Gaussian controls

The exact controls use only rational arithmetic and elementary remainder bounds. They do not share the centered-moment evaluator's recurrence. For $0 < x < 1$, let

$$A_N(x) = \sum_{j=0}^{N-1} \frac{(-1)^j x^{2j+1}}{2j+1}.$$

The alternating-series estimate gives

$$\arctan x \in \begin{cases} [A_N(x), A_N(x) + x^{2N+1}/(2N+1)], & N \text{ even,} \\ [A_N(x) - x^{2N+1}/(2N+1), A_N(x)], & N \text{ odd.} \end{cases}$$

The identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239) \tag{A.1}$$

is verified by tangent addition: if $a = \arctan(1/5)$, then $\tan(4a) = 120/119$, and $\tan(4a - \arctan(1/239)) = 1$. Elementary bounds put the angle in $(0, \pi/2)$, so it equals $\pi/4$ and not another tangent branch. The implementation uses $N = 120$ and $N = 40$ for the two arctangent series.

For the logarithm, set

$$B_N = 2 \sum_{j=0}^{N-1} \frac{(1/3)^{2j+1}}{2j+1}.$$

Integrating the geometric series for $2/(1-x^2)$, or using the power series of $2 \operatorname{arctanh} x$, gives

$$B_N < \log 2 < B_N + \frac{2(1/3)^{2N+1}}{(2N+1)(1-1/9)}. \quad (\text{A.2})$$

The code uses $N = 160$. Both interval endpoints are exact rational numbers. Cartesian complex interval multiplication then encloses $(-\frac{1}{2} \log 2 + \frac{i\pi}{4})^d/d!$.

For each depth $1 \leq d \leq 8$, the independently generated real and imaginary intervals lie wholly inside the corresponding coordinate intervals returned by the centered evaluator with 240 terms. This is an exact containment check, not a decimal comparison. It is a strong normalization and branch control, while the identity (7.1) still rests on its differential-equation proof.

The exact evaluator also supplies a dyadic upper bound on $\sqrt{x}$ for a nonnegative rational $x = a/b$. With an integer precision p , set

$$q = \frac{\lfloor \sqrt{[2^{2p}a/b]} \rfloor + 1}{2^p}.$$

Then $q^2 > x$. This is computed by integer square root and exact comparison. If its use in the generic geometric-tail formula fails to produce an upper bound smaller than one for $|w|$, the code stops rather than accepting a possibly divergent bound.

B Dependency map and integration instructions

The logical dependencies are short enough to audit separately from the implementation:

Result	Dependencies
Moment construction	Gamma integral; Fubini; the two operator identities.
Exact density zeros	Initial zero moments; Rolle's theorem; endpoint value of Af .
Positive factorization	Exact density zeros; monic sign polynomial; polynomial cancellation.
Minimality	Uniqueness of finite Hausdorff moments; sign changes.
Global nonvanishing	Positive transform signs; compensator roots lie on the removed cut.
Angular upper bound	Strict monotonicity of $t(u)$; confluent partial fractions.
Angular lower bound	Positive-transform sector; continuous unwrapped phase.
Strict radial motion	Rational positive-transform interlacing; weak approximation; simple upward phase crossings.
Exact numerical bounds	Norm bound; centered geometric series; rational arithmetic.

In particular, the numerical checks are downstream of the analytic proofs, and the angular upper and lower bounds have separate arguments.

A suggested report destination is

`Analysis/Polylogarithms/docs/reports/all-depth-signed-oscillation/`

The integration synopsis in `integration/all_depth_synopsis.tex` is a short standalone section, using prefixed macros and labels. It summarizes the proved scope and the proof chain; the full proofs remain in this article. It should be placed after the existing signed-kernel and all-positive depth-two developments, with the report cited as a separate contribution. The optional notation patch is independent of this addition.

The manuscript's editorial ledger should distinguish four things: ordinary proved theorems in this article, finite exact replay evidence, proposed further questions, and previously open arithmetic candidates whose

status is unchanged. The word “minimal” should always retain its qualifier “polynomial compensation degree” when discussing Theorem 4.3.

The package has no new dependency on a proof assistant, Wolfram Language, or the full ProveIt checkout. Python’s standard library suffices for the exact evaluator. The replay additionally uses SymPy for the stated finite symbolic checks; regenerating candidate angular brackets uses mpmath and NumPy. The README records the versions actually used. A successful rebuild may change PDF metadata; the package hash manifest describes the delivered files, not every possible future rebuild.

References

- [1] ProveIt repository, *Polylogarithms manuscript*, Chapter 5, source `chapters/05-signed-kernels.tex`, especially `signed:thm:kernel` and `signed:thm:unique-zero`. Repository snapshot 3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f, October 2026. Base directory: <https://github.com/VladimirReshetnikov/ProveIt/tree/3447bc59b78d138a7f0d536b66ac6e5cc5d5e49f/Analysis/Polylogarithms/docs/manuscript>.
- [2] ProveIt repository, same manuscript and snapshot, `chapters/05-real-orders.tex` and `chapters/05-real-local-radius.tex`. Sources for the finite signed-measure parameter domain and the local normalized-radius results.
- [3] ProveIt repository, same manuscript and snapshot, `chapters/05-subcritical-stieltjes.tex`. Sources for positive consecutive differences, all-positive depth-two nonvanishing, and subcritical radial geometry.
- [4] ProveIt repository, same manuscript and snapshot, `chapters/05-real-positive-kernel.tex`. Sources for the positive double kernel and Hurwitz–polylogarithm transport. Blob SHA 044d3825b90dce437ac9007733e6447cf9563098. The two notation replacements proposed here concern lines 102 and 104.
- [5] ProveIt repository, same manuscript and snapshot, `README.md` and `chapters/10-discovery.tex`. Sources for proof-status distinctions, remaining arithmetic candidates, and the surviving research programme. Consulted October 10, 2026.
- [6] J.-G. Liu and R. L. Pego, *On generating functions of Hausdorff moment sequences*, Transactions of the American Mathematical Society **368** (2016), 8499–8518, doi:10.1090/tran/6618. The corrected author version arXiv:1401.8052v4, November 12, 2025, including its corrigendum, was consulted: <https://arxiv.org/html/1401.8052v4>.
- [7] Jiangtao Li, *The derived set of multiple star polylogarithms*, arXiv:2609.03653v1, September 3, 2026. <https://arxiv.org/html/2609.03653v1>. This concerns weak/star sums; it is cited for comparison, not used to replace the strict-sum proof.
- [8] NIST Digital Library of Mathematical Functions, §25.12, *Polylogarithms*, and §5.5, *Functional Relations* for the Gamma function. <https://dlmf.nist.gov/25.12>; <https://dlmf.nist.gov/5.5>. Consulted October 10, 2026.